Multilevel Mapping From Remote Sensing Images: A Case Study of Urban Buildings

Yilang Shen, Tinghua Ai, Hao Chen, and Jingzhong Li

***Abstract*— Remote sensing mapping plays an important role in understanding regional development and geographical environment characteristics. Traditional remote sensing mapping at different levels usually fails to consider the shape, quantity, distribution, and position features of map objects. Therefore, a multilevel representation of urban buildings is realized based on the proposed framework for multilevel mapping from remote sensing images. In this process, the Mask R-CNN method is first applied to extract buildings from remote sensing images. Then, the orthogonal shape features of the extracted buildings are reconstructed based on corner detection, and urban roads are generated by extracting the internal structural characteristics of urban buildings for further multilevel representation. Finally, three innovative raster-based generalization algorithms, including simplification, aggregation, and typification based on Hough line detection technology, are developed for a multilevel representation of urban buildings. The experimental results reveal that the proposed methods can effectively realize multilevel mapping of urban buildings from remote sensing images while meeting basic cartographic requirements.**

***Index Terms*— Hough line detection, Mask R-CNN, multilevel mapping, remote sensing images, urban buildings.**

I. Introduction

REMOTE sensing images contribute greatly to the mapping of geographic spatial information, such as remote sensing image maps, by providing source data and background for ordinary vector maps, and remote sensing thematic maps [1]. To achieve these goals, numerous studies have focused on remote sensing image classification [2], [3], fusion [4], [5], restoration [6], [7], correction [8], [9], and change detection [10], [11]. However, the mapping scale is a basic problem that cannot be ignored during remote sensing image mapping because as the map shrinks, increasingly more details of the map objects cannot be clearly recognized. The geographic scale is important because it provides a definition of the limits of our earth observations and can serve as a parameter for shaping geographic phenomena in social and physical processes [12]. In remote sensing studies, the scale also influences many issues such as information transfer, data aggregation, and landscape analysis [13].

On the one hand, although remote sensing images can traditionally be used to generate maps at various scales, the representation scale of the map objects is limited to spatial resolutions related to sensors used and the height of flight. Methods of generating maps from remote sensing images do not consider basic cartographic characteristics considered in traditional map generalization, including shape features such as the orthogonal characteristics of buildings [14], quantity features [15], [16], distribution features such as distribution density, center, and range [17], [18] and position characteristics, such as overlays and conflicts of geographical elements between map objects [19] and Gestalt principles [20]. In other words, multilevel mapping from remote sensing images that includes multilevel representations at the pixel, target, and landscape levels is a problem that has never been identified and is far from being resolved. On the other hand, traditional cartography generalization, which aims to simplify the representation of map elements at different levels of detail, includes various generalization operations, such as simplification [21], [22], selection [23], [24], aggregation [25], collapse [26], typification [27], smoothing [28], and displacement [29], [30]. In this process, many characteristics, such as the shape, quantity, distribution, and position features of map elements, are carefully considered. For example, the generalization of urban buildings is closely linked with the subsistence of humans. When the map scale decreases, the legibility of buildings must be ensured using simplification operations that can preserve the orthogonal characteristics, and the overlays between buildings can be eliminated using aggregation and agglomeration operations, while the representation conflicts between buildings and roads can be resolved using displacement operations; moreover, distribution patterns can be preserved using the typification operation. However, most of these generalization algorithms were developed for vector data that have point, line, or area coordinate structures. Although a small number of raster-based generalization algorithms, such as mathematical morphologic [31], [32] and image segmentation [14], [21], [25] methods, have been proposed, they can only be applied to regular binary images generated from vector data with regular structures rather than remote sensing

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images with a large number of different pixel values and noise levels. Furthermore, traditional multilevel mapping of spatial data based on remote sensing images is usually divided into several separate processes, which mainly include multiresolution image processing, vectorization of raster images, and map generalization. This division of tasks can result in a relatively long production period, complex operational processes, and large errors in accuracy. Therefore, there is an urgent need to simplify these separate mapping processes to promote their integration and streamline the task of multilevel mapping from remote sensing images.

Therefore, in this article, combining traditional cartography generalization and modern computer vision technologies, we attempt to directly generate multilevel maps from remote sensing images considering multiple cartographic features. A conceptual framework for multilevel mapping from remote sensing images is proposed, and a multilevel cartographic representation of urban buildings in remote sensing images is realized based on the proposed framework.

II. Related Works

To realize multilevel mapping of urban buildings, the first step is to extract buildings using image classification. At present, the remote sensing image classification technologies [2], [3], [33] are abundant and mature. For example, Wu and Prasad [33] proposed a method for remote sensing image classification using semisupervised deep learning, which performs better than state-of-the-art semisupervised and supervised learning methods of remote sensing image classification. However, it is extremely difficult to preserve the regularity of boundaries during this process. The boundaries of extracted buildings tend to be ambiguous, and the orthogonal features of buildings cannot be well maintained.

In the process of multilevel mapping of urban buildings, simplification, aggregation, and typification are three basic generalization operations used to decrease the complexity of building objects at large scales while maintaining their primary shape, quantity, distribution, and position characteristics. According to types of spatial data structure, building multilevel representation methods are divided into raster-based and vector-based methods. Vector data apply the x- and y-coordinates to represent spatial information, which mainly includes three types of points, lines, and polygons. However, a raster consists of a pixel matrix with regular columns and rows, and each pixel contains a value used to represent spatial information.

For building simplification, the vector-based methods include the edge-move method [34], constrained Delaunay triangulation method [35], least squares method [36], [37], and optimization strategy method [15], and the raster-based methods have mainly concentrated on neural networks [38], morphological [39], [40], and superpixel segmentation [14]. For example, Damen *et al.* [39] applied morphological operators, such as opening and closure, to building simplification. This method shows that the order of opening and closure operators affects the simplification results. Meijers [40] applied the offset curves obtained from the straight skeleton to building simplification. This method can be used to simplify and amalgamate buildings into built-up areas. Cheng *et al.* [38] proposed an approach to building simplification based on a back-propagation neural network (BPNN) model. The simplification results can be effectively obtained from the BPNN model that can learn the cartographical knowledge. In addition, the ArcGIS (ESRI, Redlands, CA, USA) building simplification method is the most popular method and is widely used for vector building simplification.

For building aggregation, Delaunay triangulation technology is widely used for the aggregation of vector data [41]–[45], whereas the mathematical morphology (MMA) [32], [46] and neural networks [47] are usually used for the aggregation of raster data. For example, Chaudhry and Mackaness [43] developed a method for aggregating buildings with a simple buffering operation. Ai and Zhang [44] proposed a progressive approach for building aggregation based on the Delaunay triangulation skeleton. In this method, three important issues including conflict detection, displacements, and combinations are well-considered. Su *et al.* [32] applied the morphological operators to area features aggregation. They divided the aggregation process into two subprocesses: a combination and a shape refinement. This method has no obvious advantages compared with other methods. In addition, Shen *et al.* [25] proposed a superpixel-based method for building aggregation. A superpixel is a collection of neighboring normal pixels with similar characteristics such as color, texture, and intensity [14]. This superpixel-based method can well maintain the orthogonal characteristics but cannot preserve the curved characteristics of buildings.

For building typification, the existing methods are all based on vector data, which includes mesh [17], [27], [48], Gestalt theory [49], graph [50], [51], self-organizing maps [18], and geometric [52], [53] methods. Raster-based building typification methods have received little attention. For example, Burghardt and Cecconi [27] developed a mesh-based approach to typify buildings based on Delaunay triangulation with a high speed, which can be used for real-time applications. In this method, building typification contains two subprocesses: positioning and representation. This mesh-based building typification method can effectively maintain the density and geometrical characteristics. However, the authors tested this method only on buildings with irregular patterns; buildings with grid and linear patterns were not considered. Subsequently, Gong and Wu [48] proposed a method for building with grid patterns using Delaunay triangulation, and Wang and Burghardt [17] proposed a method for building with linear patterns using Delaunay triangulation. These authors considered only a single distribution pattern of buildings. Using the Gestalt theory, Regnauld [49] proposed a building typification method based on the minimum spanning tree technology. In this method, the authors considered the preservation of various characteristics, such as size, shape, and density. This method can effectively preserve the original distribution characteristics of buildings but buildings with grid patterns were not considered.

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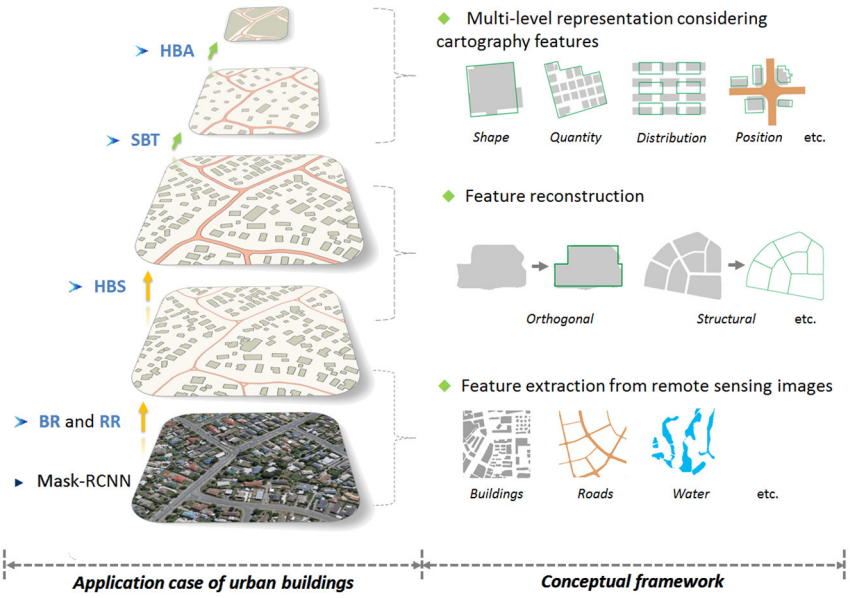


Fig. 1. Framework for multilevel mapping from remote sensing images.

From the above analysis, we observe the following three gaps in methods for multilevel mapping of urban buildings from remote sensing images.

1) There is no complete technical framework for multilevel mapping of remote sensing images. During the mapping process, the boundaries of extracted buildings are usually ambiguous. Thus, the orthogonal features of buildings need to be recovered for further cartography.
2) Compared with multilevel mapping methods based on vector data, only a limited number of methods are based on raster data. The existing raster-based building multilevel mapping methods have obvious drawbacks. For example, the raster-based building aggregation cannot simultaneously preserve the orthogonal and curved characteristics of buildings.
3) Some raster-based building multilevel mapping methods do not exist and there is an urgent need to develop them. For example, the raster-based building typification method is lacking. In addition, the existing vector-based method cannot simultaneously handle buildings with different distribution patterns such as irregular, linear, and grid patterns.

As the object extraction technologies from images are already very mature, in this study, we mainly focus on solving the above three gaps, namely, building feature reconstruction and multi-level mapping of buildings from remote sensing images.

III. Materials and Methods

A. Framework for Multilevel Mapping From Remote Sensing Images

Fig. 1 shows the proposed conceptual framework of multilevel cartography of remote sensing images. First, different geographical elements, such as buildings, roads, water, and other features, can be extracted from remote sensing images using modern computer vision technologies. With the rapid development of deep learning technologies, object detection methods from images, such as Mask R-CNN [54], have become increasingly accurate and make it possible to realize high-precision extraction. Then, a feature reconstruction process should be performed to ensure the data quality for multilevel cartography. Although high-resolution remote sensing images can be classified with a high precision, they inevitably exhibit a lack of information and contain noise. For example, the boundaries of extracted buildings may be approximate, and linear features need to be recovered for further cartography. Finally, multilevel representation based on the pixel structure should be realized by considering different cartography constraints. In this process, various constraint characteristics and criteria, such as shape, quantity, distribution, and position features, should be suitably considered by implementing traditional cartography generalization theories. Different generalization algorithms based on the pixel structure, such as simplification, aggregation, and typification, should be developed to generate multilevel representation maps of geographical elements. In other words, this is an integration process of pixel-based multilevel map generation and representation using high-resolution remote sensing images combining modern computer vision technologies and traditional cartography generalization theories. As shown on the left side of Fig. 1, an application case of multilevel mapping of urban buildings is realized to support the proposed conceptual framework. In this process, we first applied the existing Mask R-CNN model to perform a feature extraction of urban buildings. The building reconstruction (BR) and road reconstruction (RR) methods are then proposed to generate the reconstructed buildings and roads. Finally, the Hough building simplification (HBS), Hough building aggregation (HBA), and superpixel building typification (SBT) methods are designed for a multilevel representation of urban buildings.

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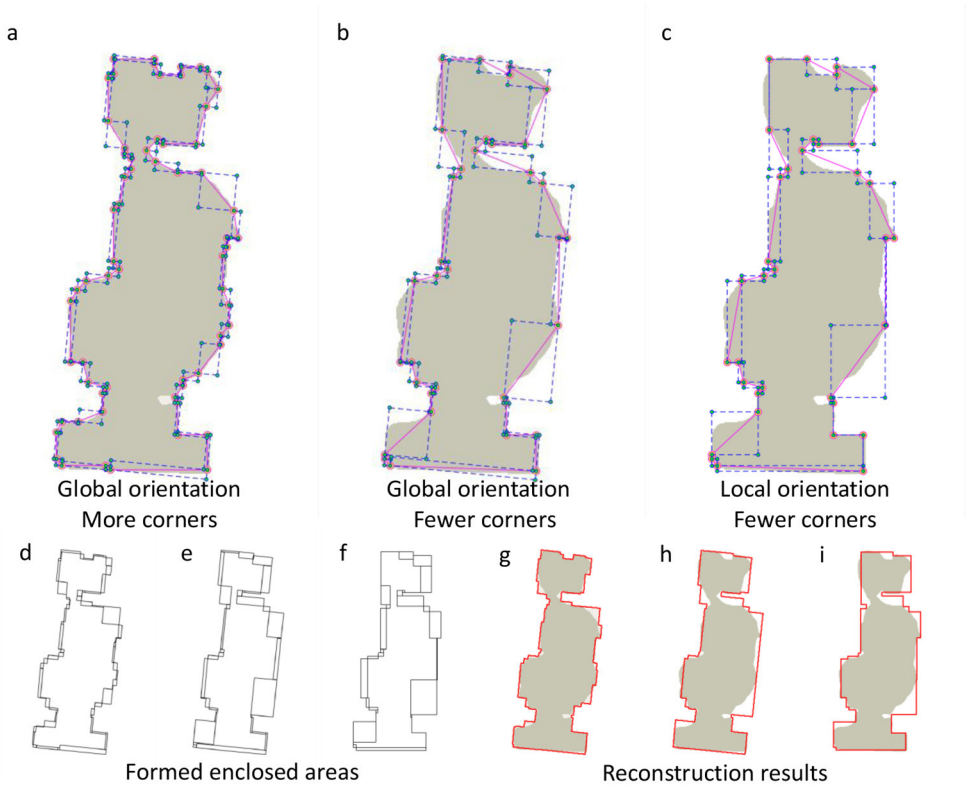


Fig. 2. Building reconstruction based on corner detection. (a) Reconstruction using the global main orientation with more corners. (b) Reconstruction using the global main orientation with fewer corners. (c) Reconstruction using the local main orientation with fewer corners. (d)–(f) Corresponding enclosed areas. (g)–(i) Corresponding reconstruction results.

B. Feature Extraction of Buildings

The rapid development of deep learning technologies provides powerful tools for the high-precision classification of remote sensing images. A modern method called Mask R-CNN [54] is applied to building extraction from high-resolution remote sensing images. Mask R-CNN is an extension of Faster R-CNN [55] that contains a new function to predict segmentation masks obtained from the original classification function in regions of interest. Mask R-CNN is easy to implement and can effectively realize object detection from images, and high-accuracy segmentation results can be represented by masks.

C. Feature Reconstruction of Buildings and Roads

1) Building Reconstruction: As the boundaries of extracted buildings are rough, a new BR method based on corner detection for the flatness of boundaries is proposed as follows. First, corner detection of buildings is conducted using the method proposed by He and Yung [56]. Second, the angle formed by three consecutive corners is calculated, and if it is close to a flat angle, then the middle one of these consecutive corners should be removed. Third, the global and local main orientations of each building are calculated. The global main orientation equals the angle between the x-axis and the long axis of the ellipse that has the same second moments as the building area. The local main orientation equals the direction of the longest line segment connected by two adjacent and approximately right corners [14]. Fourth, two orthogonal lines at each corner are determined, and the directions of these two lines are the same as the main orientation. For three consecutive corners, two line segments are drawn perpendicular to the orthogonal lines at the middle corner from the first corner, and two line segments are plotted perpendicular to the orthogonal lines at the middle corner from the last corner. The enclosed area generated by all the drawing line segments is the BR result. Fifth, the two reconstruction polygons of each building are calculated according to the global and local main orientations. The reconstruction polygon with a smaller area difference relative to the original building is the final reconstruction result.

An example of BR is shown in Fig. 2. In Fig. 2(a)–(c), the gray polygons are the original extracted buildings, and the corners are marked with larger circles, while the endpoints of the vertical line segments are marked with smaller circles. Fig. 2(a) and (b) shows the reconstruction results using the same global main orientation with different corner numbers. Fig. 2(b) and (c) depicts the reconstruction results using the same corner numbers with different global and local main orientations. Fig. 2(d)–(f) shows the corresponding enclosed areas built by all the drawing line segments. Fig. 2(g)–(i) reveals that the corresponding BR results overlap the original extracted buildings from the remote sensing images, from which it is evident that more corners can help to generate a building with more details. Since the reconstruction polygon in Fig. 2(h) has smaller area differences than the reconstruction

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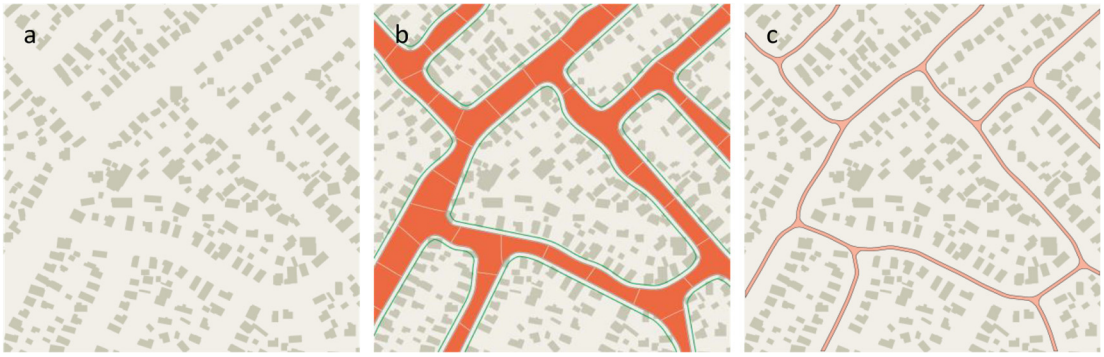


Fig. 3. Road generation from urban building blocks. (a) Original buildings. (b) Building aggregation results. (c) Road reconstruction results.

polygon in Fig. 2(i), the reconstruction polygon in Fig. 2(h) is the final result.

2) Road Reconstruction: In urban areas, buildings are usually distributed along both sides of streets. In other words, urban roads reflect the internal structural characteristics of urban buildings. Thus, a process of RR from urban building blocks combining building aggregation and centerline extraction is performed as follows.

1) Urban buildings are aggregated using the proposed method based on Hough line detection for building aggregation (HBA) as described in the section on multilevel representations of urban buildings.
2) The centerlines of urban roads are generated using the method proposed by Shen *et al.* [57].
3) A median filtering operation can optionally be applied to the generated centerlines to smooth road intersections.

An example is presented in Fig. 3. Fig. 3(a) shows the original buildings. Fig. 3(b) depicts the building aggregation results marked with blue lines and the generated red superpixels used for RR. The final RR result marked with red is shown in Fig. 3(c).

D. Multilevel Representation of Urban Buildings

1) Building Simplification: Building simplification is a process of preserving the main features while removing the redundant details of buildings [15], [36]. Certain basic geometrical characteristics, such as orthogonal shape, stair patterns, and areas, should be considered during building simplification. A new method based on Hough line detection [58] for building simplification (HBS) is proposed and consists of the following steps. First, linear features of buildings are identified using Hough line detection. In this step, the corresponding detection parameters should be reasonably adjusted to obtain relatively long line segments, and close line segments should be merged so that the original building information can be simplified. Fig. 4(a) shows that the Hough line detection result is not good. Second, if two intersecting line segments exist, the shorter line segment is removed. Third, line segments are extended to straight lines to form closed rectangular blocks, assuming that R is the building area ratio in a closed rectangular block, S is the area threshold of a building block, and N is the number of rectangular edges that overlie the building boundary. Fig. 4(b) shows a reasonable detection result; the red lines are the detected line segments, and the blue lines are the extended straight lines. Fourth, simplification rules are applied by which building parts located in open areas are removed, such as the region marked with a blue circle in Fig. 4(c). Rectangular blocks are filled when R is higher than a fixed value, such as 0.5, and conversely, the building parts located in these rectangular blocks are removed, such as the region marked with the blue ellipse in Fig. 4(c). Blank rectangular blocks are filled when $N = 2$, and S is smaller than a fixed value, as shown in Fig. 4(d). Blank rectangular blocks are filled when $N = 3$, and S is smaller than a fixed value, as shown in Fig. 4(f). Fifth, area preservation (optional): when all rectangular blocks are filled, the simplified building is resized until its area equals the original area using dilation and erosion operations. Sixth, the adjustment in the main direction: to ensure the accurate orthogonality of the simplified building boundaries, usually, the direction of each line segment in step 3 should be first adjusted to be the same as the local main orientation or perpendicular to the local main orientation of this building.

2) Building Aggregation: Building aggregation can help to decrease the number of buildings while preserving the distribution range [41], [42]. A new method based on Hough line detection and superpixel segmentation for building aggregation (HBA) is proposed as follows. First, buildings are clustered using the method proposed by Shen *et al.* [25], and Fig. 5(b) shows an example of the cluster result from Fig. 5(a). Second, bending characteristics of boundaries are acquired using median filtering, as shown in Fig. 5(c). Third, linear features of boundaries are obtained using Hough line detection. In this step, the corresponding detection parameters should be reasonably adjusted to obtain the relatively long line segments, and closed line segments should be merged. If two intersecting line segments exist and the angle between these two line segments is smaller than a fixed threshold, the shorter line segment should be removed, as shown in Fig. 5(d) and (e), where the green line segments are the shorter line segments that should be removed. Fig. 5(e) shows an example with fewer linear features. Fourth, the bending characteristics and linear features of boundaries are integrated by filling the relatively small areas of the holes between the Hough line segments and boundaries after median filtering and removing

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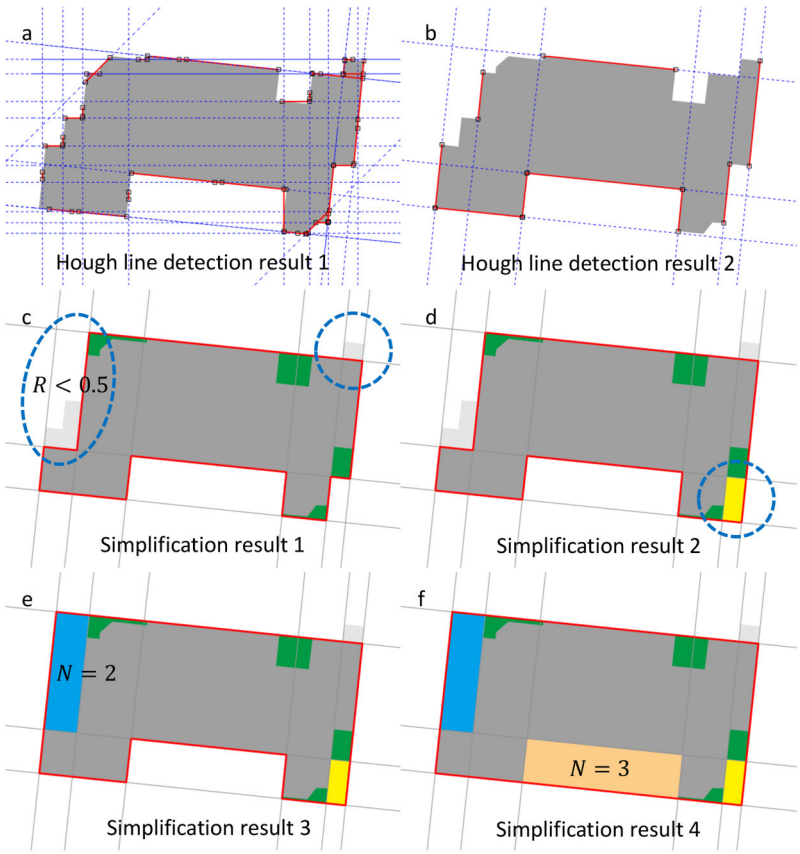


Fig. 4. Building simplification based on Hough line detection. (a) Hough detection results with redundant lines. (b) Hough detection results with suitable lines. (c)–(f) Simplification results at different levels of detail.

the relatively small areas of the redundant parts located outside the Hough line segments. Fifth, step 2 is optionally repeated to smooth joints between bending and linear characteristics. In this process, orthogonal features can still be preserved by overlap analysis of the aggregation regions before and after smoothing. The final building aggregation result is shown in Fig. 5(f). Compared with traditional building aggregation methods that can only preserve orthogonal characteristics [25], the proposed HBA method can effectively preserve bending, linear, and orthogonal characteristics at the same time.

3) Building Typification: Building typification is used to preserve the distribution characteristics while reducing the number of buildings [17], [27]. Traditional building typification methods are designed for vector data. In this article, a new SBT method combining mathematical morphologies is developed for image data according to the following steps. First, using the proposed HBA method, the typification areas of building groups are structured, as shown in Fig. 6(b), where the typification area of the original building groups is marked with light gray. Second, superpixels are generated based on the LSC method [59]. The LSC method has a good ability to preserve the original building boundaries, and the segmentation result marked in blue is shown in Fig. 6(b). Third, custom processing of superpixels is sequentially performed as follows. When a small superpixel is completely located inside another large superpixel, this small superpixel is removed. When a building is divided by more than one superpixel, these superpixels are merged. When a superpixel boundary coincides with a building boundary, this superpixel should be merged into the adjacent superpixel with the longest overlapping boundary. Fourth, building typification; the elongation and orientation of typified rectangular buildings are the same as those of the maximum-area building in each LSC superpixel. The area of typified buildings can be dynamically adjusted using dilation and erosion operations with a rectangular structural element. The location of typified buildings coincides with the center of mass of corresponding LSC superpixels, such as the red point shown in Fig. 6(b). Fig. 6(c) depicts the typified results of this example.

IV. Experiments and Evaluations

A. Feature Extraction

We initialize the Mask R-CNN model used for automatic building detection by applying the weights obtained from pretraining on the Common Objects in Context (COCO) data set [60]. COCO provides a total of 2.5 million labeled instances in 328k images, which is the most comprehensive data set of semantic detection so far. By using the method of transfer learning and retraining of the building sample data set on the pretrained model, the model accuracy and performance can be effectively improved.

Facing the task of automatic building detection, we use 4736 images that contain 130.500 buildings from the Wuhan

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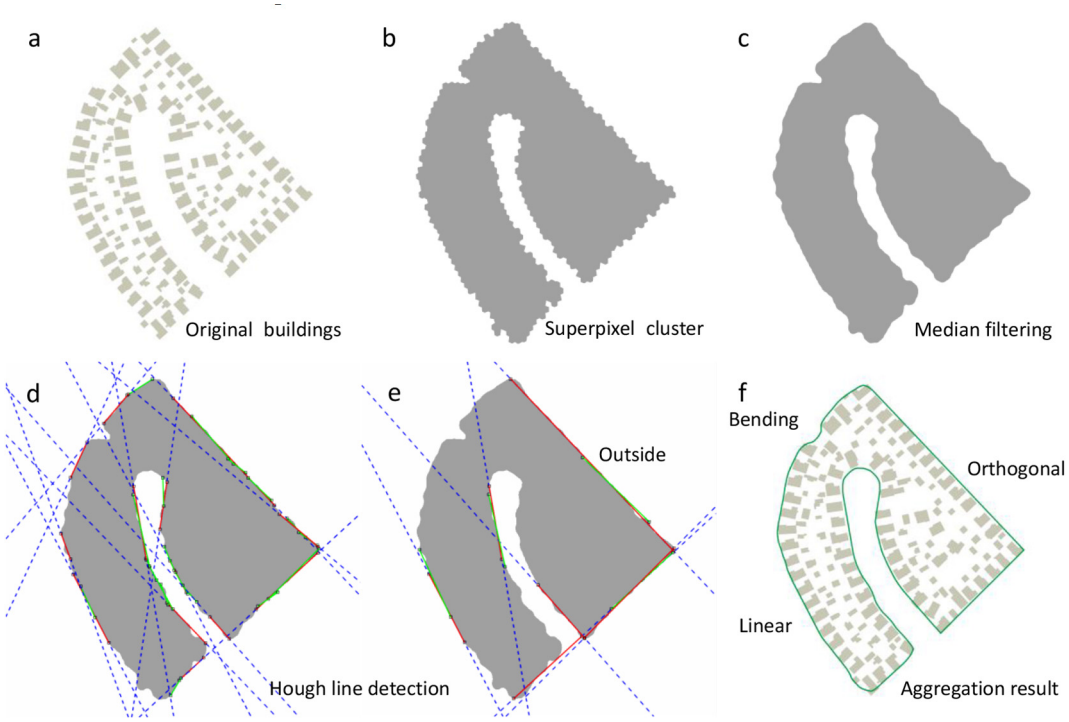


Fig. 5. Building aggregation based on Hough line detection. (a) Original buildings. (b) Clustered buildings using superpixel segmentation. (c) Median filtering results. (d) Hough detection results with redundant lines. (e) Hough detection results with suitable lines. (f) Building aggregation results.



Fig. 6. Building typification using linear spectral clustering segmentation. (a) Original buildings. (b) LSC superpixel segmentation of the typification areas. (c) Building typification results.

University (WHU), Wuhan, China, building data set [61] as the training set. In total, we train for 500 epochs using the stochastic gradient descent method with a momentum of 0.9, starting at a learning rate of 0.003 and ending at a learning rate of 0.0001. We train on a single NVIDIA GeForce RTX 2080 GPU and set the batch size as 4. Although we trained for longer times or at other learning rates, the loss function of the model did not attain additional improvements.

To quantitatively evaluate the trained model, we select 1416 images from the WHU building data set as the validation set. We use the mean average precision (mAP), the mean of the average precision (AP) in multiple categories, as an indicator of the accuracy of the model [60]. In the field of object detection, a precision and recall (P–R) curve can be plotted for each category according to the P–R, and AP is the area under the P–R curve. However, in this study, the building is the only category in the model, so the value of mAP is consistent with the AP value of the building. Through experimentation with the validation set, we obtain mAP $= 0.8010$. Basically, all buildings can be identified in the images, but their outlines are somewhat uneven.

Fig. 7(a) shows the original remote sensing images of the experimental area, and Fig. 7(b) shows the building extraction results using the Mask R-CNN model, from which we can see that although the buildings can be extracted from the remote sensing images with a relatively high precision, the linear and orthogonal information of the vast majority of the building boundaries are lost, an example of which is shown in the enlarged display in the lower right corner. There are a total of 2073 buildings in this region. The statistical geometric information, including the number, area, and perimeter of the buildings in this region, is listed in Table I.

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Fig. 7. Original remote sensing images and building extraction results. (a) Original remote sensing images. (b) Building extraction results using the Mask R-CNN model.

B. Feature Reconstruction

The proposed BR method based on corner detection is applied to recover the linear and orthogonal information of the building boundaries. The parameter values used for corner detection are $c = 1.0$, $s = 3$, $h = 0.35$, $l = 0$, and $g = 1$, where c is the minimum ratio between the long and short axes of an ellipse, s is the standard deviation of the Gaussian filter, h and l are the high and low values, respectively, used for Canny

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Fig. 8. Building and RR results and evaluations. (a) Building and RR results. (b)–(d) Typical regions of the reconstruction results. (e)–(h) Frequencies of the area, length, and centroid changes, and the sd changes in the buildings after reconstruction.

edge detection, and g is the value to fill the contour gaps. The choice of these parameters is based on the default values recommended by He and Yung [56] and has proven to be more robust in reducing the number of false corners detected compared with other traditional corner detection methods. This is beneficial for improving the accuracy of building feature reconstruction.

Fig. 8(a) shows the BR result marked with blue solid lines, and the original extracted buildings are marked with a gray color. Fig. 8(b)–(d) shows the detailed information in the

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TABLE I
QUANTITATIVE, ATTRIBUTIVE, POSITIONAL, AND SHAPE VALUES OF THE BUILDING EXTRACTION AND RECONSTRUCTION RESULTS

	Number	Total area	Average area	Total length	Average length	Mean average precision
Building extraction	2073	7222219	3483.95	498850.73	240.64	0.801
	Average area change		Average length change	Average centroid change		Average surface distance
Building reconstruction	0.098		0.063	2.526		0.163

corresponding regions marked in Fig. 8(a), from which we can see that the irregular boundaries of the originally extracted buildings are converted into linear and orthogonal boundaries; typical examples are marked with red ellipses. However, as BR is a process of feature fitting, it is difficult to avoid producing differences. Usually, feature reconstruction produces three types of changes, namely, attributive, positional, and shape changes. Thus, in this study, the area and length changes are used to evaluate the attributive changes, the centroid change is used to evaluate the positional changes, and the surface distance (sd) [14], which measures the shape similarity of two polygons, is used to evaluate the shape changes. The sd value of two polygons is calculated as follows:

$$\mathrm{sd} = 1 - \frac{\mathrm{area}(px \cap y)}{\mathrm{area}(px \cup py)}. \tag{1}$$

In this equation, $\mathrm{area}(px \cap py)$ denotes the area of the intersection of polygons px and py, and $\mathrm{area}(px \cup py)$ denotes the area of the union of these two polygons.

Fig. 8(e)–(h) shows the frequency distribution histograms of the area, length, and centroid changes and the sd changes in the buildings after reconstruction, from which we can see that most of the area and length change values are approximately 0.12, and the absolute value of the area and length change ratio is basically less than 0.5. Most of the centroid change values are approximately 2 and less than 4. Most of the sd values are approximately 0.08 and less than 0.2. The corresponding average area, length, and centroid changes and the average sd of all the buildings are summarized in Table I. The average area and length changes are less than 10% and 7% (only 0.098 and 0.063), respectively, which shows a good ability to preserve the area and length. The average centroid change is 2.526 pixels. The average sd (0.163) is far below the standard value (0.25) [14], which means that the buildings before and after reconstruction have a high degree of overlap.

The RR result of the proposed method is marked orange in Fig. 8. In this process, the superpixel size used to aggregate the urban buildings is 40 pixels, the aperture linear size of median filtering is 127, and the superpixel size used to generate the centerlines is 200 pixels. The proposed method can effectively generate road centerlines from urban buildings, and road intersections can be suitably restored, as shown in Fig. 8(a)–(d).

C. Multilevel Representation of Urban Buildings

After feature reconstruction at level 1, as shown in Fig. 9, the urban buildings are represented at different levels of detail. In this process, different building generalization algorithms considering shape, quantity, distribution, and topology features are applied to generate the representation results at levels 2–4.

From levels 1–2, the proposed HBS simplification method is applied to simplify the shape of the boundaries of the reconstructed buildings at level 1. Fig. 9(a) shows a representation of the urban buildings at level 2. The buildings at level 2 are effectively simplified, and the orthogonal features of the buildings are preserved. Compared with the classic widely used building simplification methods of ArcGIS (ESRI), Domap [22], and MWBR [63] for vector data, the proposed HBS method has the following advantages.

1) The HBS and MWBR methods generate completely orthogonal boundaries, while the ArcGIS and Domap methods tend to generate nonorthogonal boundaries in some cases, as indicated by the arrows in Fig. 10(a) and (b) and (e) and (f).
2) While the ArcGIS and Domap methods generate locally uneven boundary boundaries in some cases, the HBS and MWBR methods effectively avoid this, as indicated by the arrows in Fig. 10(c) and (d) and (g) and (h).
3) In some cases, the MWBR method generates intersecting buildings, whereas the Domap method destroys the original orthogonal characteristics, such as the regions marked with red circles in Fig. 10(e), (f), and (h). Hence, the proposed HBA method performs better in these two aspects.
4) The frequency distribution histograms of the area, length, and sd changes in the buildings after simplification using the HBS, ArcGIS, Domap, and MWBR methods are presented in Fig. 10(i)–(t).

Since different algorithms that lead to different types of building changes are applied at levels 2–4, we used different comparison indicators at these levels. Table II shows the quantitative, attributive, and shape values of building representations at levels 2–4. The sd of the proposed HBS method is only 0.147, which is slightly greater than that of the ArcGIS (0.100), MWBR (0.122), and Domap (0.142) methods, but it is still suitable compared with the standard of 0.25. The change in length of the proposed HBS method is only 6.2%, which is less than that of the MWBR (7.3%) method, but greater

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Fig. 9. Multilevel mapping results for urban buildings. (a) Building simplification results at different levels of detail. (b) Building typification results at different levels of detail. (c) Building aggregation results at different levels of detail.

than that of the ArcGIS (1.0%) and Domap (5.6%) methods. However, the change in the area (only 0.3%) produced by the HBS method is far smaller than the area changes produced by the ArcGIS (10.4%), MWBR (19.4%), and Domap (8.3%) methods, which means that the proposed method can better preserve the original area. The runtime of the ArcGIS, MWBR, Domap, and HBA methods is 3.0, 1.0, 43.5, and 1208.4 s, respectively. Compared with the vector-based simplification methods (ArcGIS, MWBR, and Domap), the proposed raster-based HBA method is more time-consuming mainly because the needed information storage space for an object of raster data is larger than that for vector data, which results in a relatively large number of computations. Using the proposed HBS method, the representation of urban buildings from levels 1 to 2 can be divided into multiple sublevels if necessary, as shown on the right side of Fig. 9(a).

From levels 2 to 3, the proposed SBT typification method is applied to decrease the number of buildings while preserving the relative distribution characteristics. Fig. 9(b) shows the typified representation of urban buildings at level 3. In this process, the superpixel size used to generate the typification areas of building groups is 40 pixels, and the superpixel size used for building typification is 150 pixels. The proposed SBT method can effectively preserve the original distribution patterns, including linear [Fig. 11(a) and (b)], grid [Fig. 11(c)], and irregular [Fig. 11(d)] patterns. The relative distribution densities of the original and typified buildings are calculated, as shown in Fig. 11(e) and (f), respectively. Yellow represents

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TABLE II
QUANTITATIVE, ATTRIBUTIVE, AND SHAPE VALUES OF THE BUILDING REPRESENTATIONS AT LEVELS 2–4

Level 2	Methods	Length change	Surface distance	Area change
	HBS	-0.062	+0.147	-0.003
	ArcGIS	+0.010	+0.100	+0.104
	MWBR	+0.073	+0.122	+0.194
	Domap	-0.056	+0.142	-0.083

Level 3	Number change	Total area change	Total length change	Average area change	Average length change
	-0.651	-0.507	-0.573	+0.412	+0.223

Level 4	Methods	Cluster number	Total area change	Total length change
	HBA	36	+2.265	-0.710
	ArcGIS	37	+1.297	-0.618
	ArcGIS-O	68	+1.252	-0.602
	SUPA	36	+2.162	-0.720
	MMA	43	+1.482	-0.673

a high density, and blue represents a low density. Fig. 11(e) and (f) reveals that although the total density is reduced during typification, the relative distribution density is preserved. For example, the density in region A is relatively high, and the buildings in region B have a medium density, while the density in region C is relatively low. The number, area, and length change information at this level are listed in Table II. Although the total area and length changes are down by approximately 50%, the average area and length changes rise by 0.412 and 0.223, respectively. This occurs because the buildings with the largest area locally are selected to be typified buildings. Using different parameter values, the typification level can be divided into multiple sublevels, as shown for levels 3a and 3b in Fig. 9(b). Combining Töpfer's radical law [62] for determining the number of objects at different levels, the equation to set the superpixel size at different typification levels uses the proposed SBT method as follows:

$$\text{Size}_S = \frac{\text{Area}_I}{\text{Num}_O \times \sqrt{\text{Scale}}}.$$

In this equation, Num_O is the number of buildings before typification, Scale is the multiple of image reduction, Area_I is the size of the typification region, and Size_S is the superpixel size. Since the number of superpixels generated from the LSC method is related to the shape and resolution of an object, it cannot generate a specified number of superpixels by setting a fixed superpixel size. Thus, the optimal superpixel size value at the corresponding level is acquired by adjusting around the value of Size_S. Accordingly, the final superpixel sizes used for building typification at levels 3a and 3b are 150 and 185 pixels, respectively.

During the multilevel representation of urban buildings from levels 1 to 3, overlap between buildings and roads may occur. To resolve any representation conflicts while preserving the original topological relationships, the following four optional rules should be applied.

1) When two buildings overlap, remove the smaller building.
2) When a building and a road overlap and the overlapping area is smaller than the area threshold of the building, the erosion operation can be applied to remove the overlap, as indicated by the blue arrow in the bottom right of Fig. 9(b).
3) When a building and a road overlap and the overlapping area is larger than the area threshold of the building, the overlapping orthogonal parts can be directly removed, as in the case of the white part of the building indicated by the orange arrow in Fig. 9(b). If the area of the remaining orthogonal parts is smaller than the area threshold of the building, remove the whole building.

From levels 3 to 4, the proposed HBA aggregation method is applied to aggregate buildings to preserve the global shape of building groups and decrease the quantity. Fig. 9(c) shows the aggregation results of buildings at level 4. The superpixel sizes used for building clustering at levels 4a and 4b are 40 and 80 pixels, respectively, and the aperture linear sizes of median filtering at these two levels to acquire the bending characteristics of boundaries are 127 and 237, respectively, while the corresponding distance between the two merged line segments used to acquire the linear characteristics of boundaries is $250 \leq d \leq 500$. During the choosing of parameter values at different aggregation levels, we recommend that the superpixel sizes be set to equal the widths of the roads at different grades, and the aperture linear sizes of median filtering be odd numbers that are approximately equal to n times ($n = 3$ in this case) greater than the superpixel sizes. We compared the proposed HBA method with four aggregation methods, the ArcGIS method preserving the original shape, the ArcGIS-O method preserving the orthogonal shape, the superpixel-based (SUPA) [25] method, and the MMA [31] method. Fig. 12 shows the differences among these five methods. As indicated by the purple arrow, the proposed HBA method better preserves the smooth aggregate boundaries, while the ArcGIS, ArcGIS-O, SUPA, and MMA aggregation methods tend to generate uneven boundaries. As indicated by

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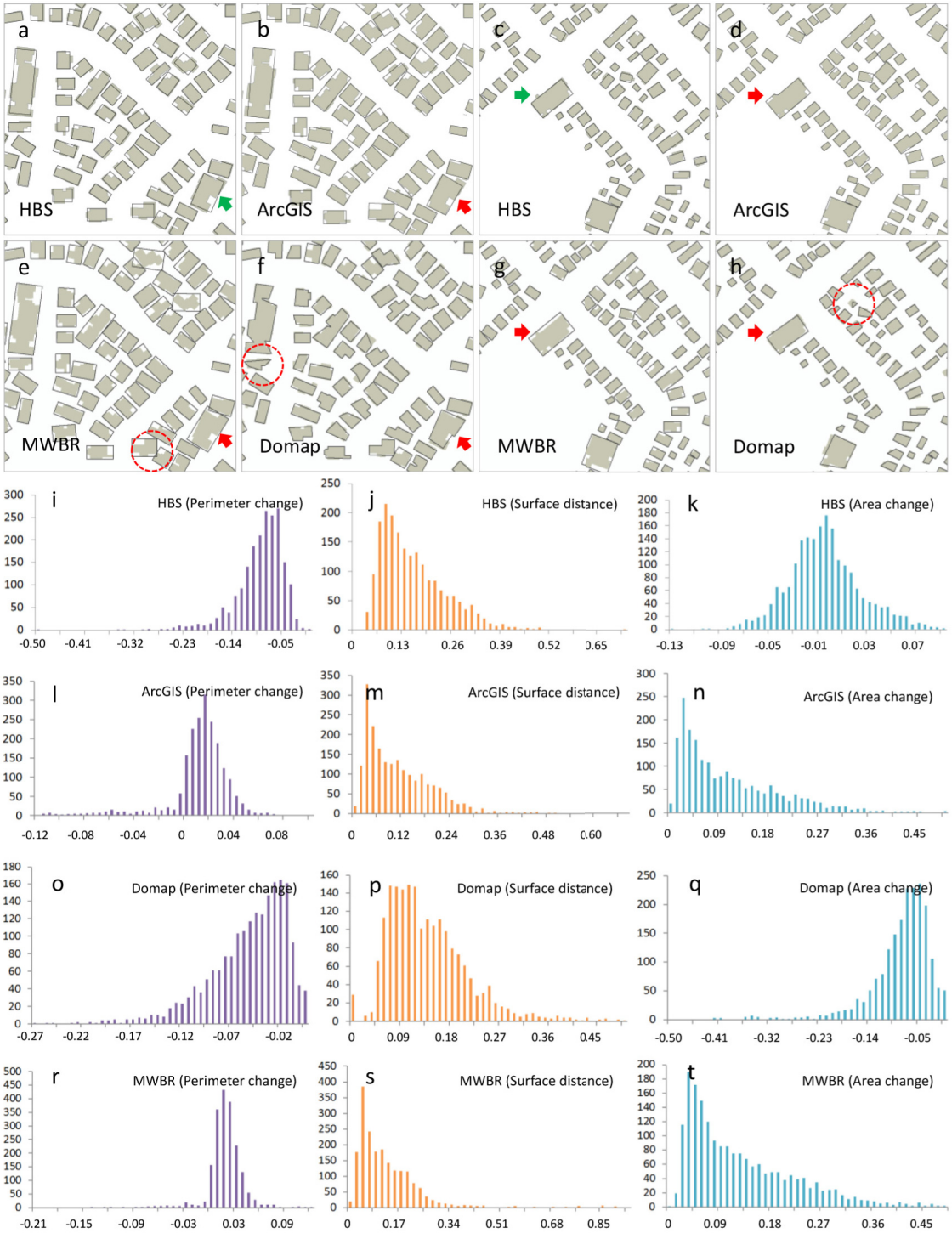


Fig. 10. Evaluations of urban building simplification. (a)–(h) Building simplification results using the HBS, ArcGIS, MWBR, and Domap methods. (i)–(t) Frequencies of the length, area, and sd changes in the buildings after simplification using the four methods.

the blue arrow, the proposed HBA method better preserves the global linear characteristics of building groups, but the ArcGIS, ArcGIS-O, SUPA, and MMA aggregation methods cannot achieve the same level of preservation. In addition, the ArcGIS-O method fails to aggregate close buildings in some cases, even when the aggregation distance is sufficiently large, as shown in the red circles, while the proposed HBA method effectively avoids this, as indicated by the green circles. Table II presents the differences in cluster numbers, total area changes, and total length changes among the five aggregation methods. After aggregation using the proposed HBA method, the cluster number is 36, the total area increases by 2.265, and the total length decreases by 0.710. The cluster numbers generated from the HBA (36), ArcGIS (37), SUPA (36), and MMA (43) are relatively close and lower than that of the ArcGIS-O method (68). This is because the ArcGIS-O

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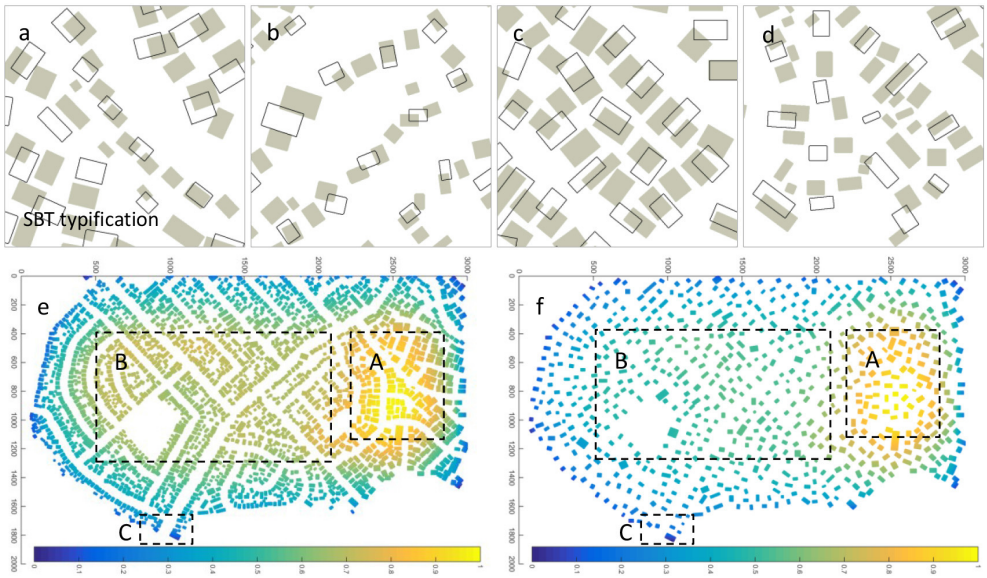


Fig. 11. Evaluations of urban building typification. (a)–(d) Typical regions in the building typification results using the proposed SBT method. (e)–(f) Densities of the buildings before and after typification.

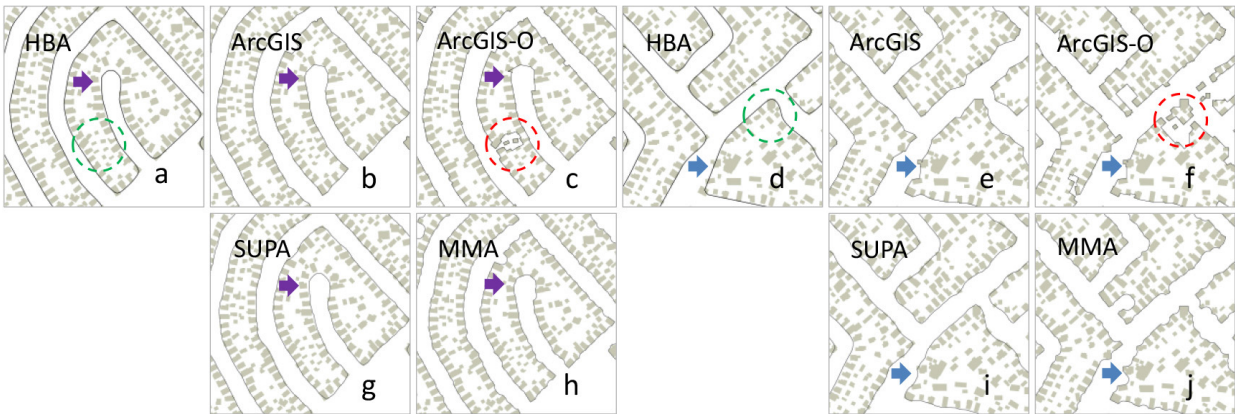


Fig. 12. Evaluations of urban building aggregation. (a)–(j) Building aggregation results using the HBA, ArcGIS, ArcGIS-O, SUPA, and MMA methods.

method fails to aggregate close buildings in some cases. The total area changes of the HBA (+2.265) and SUPA (+2.162) are relatively close but less than that of the ArcGIS (+1.297), ArcGIS-O (+1.252), and MMA (+1.482) methods. The same rule applies to the total length change. This is because the HBA and SUPA methods can fill the small holes of polygons during aggregation. The runtime of the ArcGIS, ArcGIS-O, SUPA, MMA, and HBA methods is 3.0, 3.0, 715.8, 1248.4, and 936.5 s, respectively, thus indicating that the vector-based methods (ArcGIS and ArcGIS-O) perform better than the raster-based methods (SUPA, MMA, and HBA) in computational efficiency. Using different parameter values, the aggregation level can be divided into multiple sublevels, as shown for levels 4a and 4b in Fig. 9(c).

V. Conclusion

By combining modern computer vision technology and typical cartographic theories, this study realizes the multilevel mapping of urban buildings from remote sensing images while considering the shape, quantity, distribution, and position features of the map objects. The following conclusions can be drawn.

1) The proposed method for feature reconstruction can effectively restore the orthogonal characteristics of buildings and generate road centerlines from urban buildings. In this process, the road intersections can be suitably restored.
2) The proposed HBS simplification method can be used for raster building simplification and to generate completely orthogonal boundaries, but the traditional ArcGIS method for vector building simplification tends to generate nonorthogonal and locally uneven boundaries in some cases.
3) The proposed HBA aggregation method used for raster building aggregation can simultaneously preserve smooth aggregate boundaries and global linear characteristics of building groups, while the traditional vector aggregation methods in ArcGIS cannot achieve this result and tend to generate uneven boundaries.
4) The proposed SBT typification method used for raster building typification can effectively decrease the number

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of buildings while preserving the relative distribution characteristics including distribution patterns and density. Compared with the existing vector-based typification methods, the proposed SBT can simultaneously handle buildings with different distribution patterns such as irregular, linear, and grid patterns.

In addition, compared with the multilevel mapping model, several potential advantages of the proposed framework for multilevel mapping from remote sensing images are emphasized. First, the versatility of mapping data. In traditional multilevel mapping methods, the vector data structure is complex, and there are numerous ways to categorize it, such as entity and topology type, which is not convenient for data standardization, normalization, and exchange. Thus, the data versatility of traditional mapping methods is hampered. However, the map data of the proposed framework are simple raster data with a uniform matrix structure, which has a good versatility. Second, conciseness of mapping process. Traditional multilevel mapping usually requires more human intervention, such as data format conversion, data quality correction of the vectorization of raster images. Thus, it is difficult to guarantee the accuracy of the data with respect to consistency and topological relations. However, using the proposed mapping framework, these problems can be avoided. Third, up-to-date mapping results. Remote sensing images exhibit good real time. As the proposed framework has a relatively concise data structure and mapping process, it can quickly generate up-to-date mapping results from remote sensing images compared with traditional mapping methods.

Some limitations and possibilities for further work are presented as follows.

1) This study examines only the multilevel mapping of urban buildings; in the future, the multilevel mapping of other geographical elements in remote sensing images, such as water bodies, roads, and vegetation, should be further studied.
2) The consideration of the topological relationships among urban buildings is limited here because we examine only the topological relationships between roads and buildings. Hence, the complex topological relationships among multiple geographical elements should be further studied.
3) During multilevel representation of urban buildings, additional raster-based building generalization operations such as displacement should be further developed.

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